

Ternary liquid–liquid equilibria of trifluorotris(perfluoroethyl)phosphate based ionic liquids + methanol + heptane

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ABSTRACT

Ionic liquids: (4-(2-methoxyethyl)-4-methylmorpholinium trifluorotris(perfluoroethyl)phosphate, 1-(2-methoxyethyl)-1-methylpiperidinium trifluorotris(perfluoroethyl)phosphate, 1-(2-methoxyethyl)-1-methylpyrrolidinium trifluorotris(perfluoroethyl)phosphate have been investigated in separation of heptane/methanol mixture which form an azeotrope. Three systems of ternary liquid–liquid equilibria containing mentioned ionic liquids + methanol + heptane have been determined at $T=298.15$ K. All systems show total miscibility of methanol in the ionic liquid and low solubility of heptane. The experimental results have been correlated using NRTL model. The selectivity and the solute distribution ratio have been calculated for all systems. High selectivities and solute distribution ratios have been obtained for all systems. Experimental tie-lines as well as selectivities and solute distribution ratios are compared with other ionic liquids. The influence of cation and anion structure on phase equilibria is discussed.

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1. Introduction

Many compounds form azeotropes in industrial processes. For instance, alkanes/alcohols mixtures which exist in petrochemical processes. It is caused by addition alcohols as oxygenate additive to gasoline due to reduction of lead in fuels. Separation of these mixtures is important because pure components can be reuse in further applications. Unfortunately, in the case of azeotropes this process is difficult to achieve and unrealizable using simple distillation. The most common technique is extractive distillation, however it is energetically unfavorable. Liquid–liquid extraction (LLE) method indicates lower cost and it is more friendly for environment. Thus, appropriate entrainer with good selectivity is required. Ionic liquids (ILs) can be used in this process, because they usually reveal miscibility with polar organic and inorganic compounds. There are salts with asymmetric structure generally melting in temperature below 100°C . These compounds have unique properties which allow considering them as “green” replacement for toxic and volatile solvents, namely negligible vapor pressure, wide liquid range and thermal and chemical stability. Due to large number structure of cations and anions it is possible modification physical and chemical properties of ILs for specific processes, what is additional advantage. A lot of information about extractive ability of ILs and influence of IL structure on phase equilibria give triangle

liquid–liquid equilibria in different systems. Studies of separation alkanes/alcohols [1–3] as well as other mixture [4–13] using ILs are presented in literature. Listed papers show that ILs can be used as potential entrainers in separations. In this work ternary LLE of three ILs based on trifluorotris(perfluoroethyl)phosphate $[\text{FAP}]^{-}$ anion + methanol + heptane were determined. Methanol and heptane form azeotropic mixture at 0.515 mass fraction of methanol at $T=332.3$ K [14]. Obtained results were compared with three bis(trifluoromethylsulfonyl)amide $[\text{NTf}_2]^{-}$ based ILs + methanol + heptane systems investigated in our previous work [3]. In both cases the structures of cations were the same, namely 4-(2-methoxyethyl)-4-methylmorpholinium $[\text{COC}_2\text{mMOR}]^{+}$, 1-(2-methoxyethyl)-1-methylpiperidinium $[\text{COC}_2\text{mPIP}]^{+}$ and 1-(2-methoxyethyl)-1-methylpyrrolidinium $[\text{COC}_2\text{mPYR}]^{+}$.

2. Experimental

2.1. Chemicals

The list of chemicals used in this study is presented in the Table 1 (with abbreviations of ILs). The ionic liquids were further purified by subjecting the liquid to a very low pressure of about 5×10^{-3} Pa at a temperature about 95°C for ca. 5 h. This procedure removed any volatile chemicals and water from the ionic liquid. The rest of chemicals were used without further purification due to high purity.

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Table 1

Description of chemicals.

Chemical name ^a	Source	Initial mass fraction purity	Purification method	Final mass fraction purity	Analysis method
Trifluorotris(perfluoroethyl)phosphate; 4-(2-methoxyethyl)-4-methylmorpholin-4-ium (4-(2-methoxyethyl)-4-methylmorpholinium trifluorotris(perfluoroethyl)phosphate) abbreviation: [COC ₂ mMOR][FAP]	Merck	≥0.995	Vacuum heating	—	—
Trifluorotris(perfluoroethyl)phosphate; 1-(2-methoxyethyl)-1-methylpiperidin-1-ium (1-(2-methoxyethyl)-1-methylpiperidinium trifluorotris(perfluoroethyl)phosphate) abbreviation: [COC ₂ mPIP][FAP]	Merck	≥0.995	Vacuum heating	—	—
Trifluorotris(perfluoroethyl)phosphate; 1-(2-methoxyethyl)-1-methylpyrrolidin-1-ium (1-(2-methoxyethyl)-1-methylpyrrolidinium trifluorotris(perfluoroethyl)phosphate) abbreviation: [COC ₂ mPYP][FAP]	Merck	≥0.995	Vacuum heating	—	—
Methanol	Fluka	≥0.999	—	0.9997	GC ^b
Heptane	Fluka	≥0.99	—	0.9979	GC ^b
Propan-2-one (acetone)	Fluka	≥0.999	—	0.9997	GC ^b
Butan-1-ol	Sigma-Aldrich	≥0.998	—	0.9988	GC ^b

^a Names in parentheses are common names used in text.^b Gas-liquid chromatography.**Table 2**Values of binary parameters for the NRTL equation^a for the ternary systems {IL (1) + methanol (2) + heptane (3)} at T = 298.15 K.

IL	ij	$g_{ij}/J \text{ mol}^{-1}$	$g_{ji}/J \text{ mol}^{-1}$	σ_x
[COC ₂ mMOR][FAP]	12	−11,438	12779	0.006
	13	9513	17,706	
	23	4859	3718	
[COC ₂ mPIP][FAP]	12	−10,226	6727	0.005
	13	6528	17,109	
	23	4859	3718	
[COC ₂ mPYP][FAP]	12	−9019	8705	0.006
	13	8109	21,919	
	23	4859	3718	

^a Parameter $\alpha = 0.3$.

2.2. Apparatus and procedure

The structures and properties like water contents, densities, viscosities, decomposition temperatures, temperatures of fusion, enthalpies of fusion, glass transition temperatures and the change of heat capacity at the glass transition temperatures of investigated ILs were presented in previous work [10]. The apparatus and the method of determination of experimental LLE tie-lines have been described in the previous work [3].

3. Modeling

The LLE was correlated with the NRTL model describing the excess Gibbs energy [15]. The equations were described elsewhere [12,16]. The NRTL nonrandom parameter α was set to a value of $\alpha = 0.3$, which has given the best results of the correlation. The results of the correlation, values of the model parameters, and the corresponding standard deviations, σ_x defined as:

$$\sigma_x = \left(\frac{\sum_{k=1}^N \sum_{i=1}^3 \left(x_{i,k}^{\text{IL,exp}} - x_{i,k}^{\text{IL,calc}} \right)^2 + \left(x_{i,k}^{\text{HC,exp}} - x_{i,k}^{\text{HC,calc}} \right)^2 \right)}{6N} \quad (1)$$

where N is the number of tie lines, are given in Table 2. The results of the correlation are presented in Figs. 1–3.

4. Results and discussion

To evaluate ability of ILs as solvent in LLE of heptane/methanol mixture, the selectivity S and the distribution ratios β are calculated using the following equations:

$$S = \frac{x_2^{\text{IL}} x_3^{\text{HC}}}{x_2^{\text{HC}} x_3^{\text{IL}}} \quad (2)$$

$$\beta = x_2^{\text{IL}} / x_2^{\text{HC}} \quad (3)$$

where x is the mole fraction, subscripts 2 and 3 refer to methanol and to heptane, respectively, and superscripts HC and IL indicate the hydrocarbon-rich phase and the IL-rich phase, respectively.

The selectivity informs how many stages are needed in separation processes. The solute distribution ratio means the capacity of IL and determines amount of IL required to use in extraction. High values of these parameters are desirable because it provides to smaller number of stages and less amount of IL and consequently, it leads to lower operating costs.

Table 3 lists the compositions of the experimental tie-lines for each ternary systems IL + methanol + heptane at T = 298.15 K and calculated values of S and β . Experimental and NRTL correlation LLE tie-lines are shown in the triangular diagrams in Figs. 1–3. It is

Table 3

Compositions of experimental tie lines, solute distribution ratios, β and selectivity, S for ternary systems {IL (1) + methanol (2) + heptane (3)} at $T = 298.15$ K^a

x_1^{HC} hydrocarbon-rich phase	x_2^{HC} hydrocarbon-rich phase	x_1^{IL} IL-rich phase	x_2^{IL} IL-rich phase	S	β
[COC ₂ mMOR][FAP]					
0.000	0.000	0.978	0.000	–	–
0.000	0.006	0.860	0.115	762	19.2
0.000	0.008	0.694	0.280	1335	35.0
0.000	0.009	0.598	0.377	1660	41.9
0.000	0.009	0.594	0.380	1609	42.2
0.000	0.010	0.529	0.449	2021	44.9
0.000	0.014	0.423	0.553	1623	39.5
0.000	0.019	0.359	0.617	1327	32.5
0.000	0.020	0.349	0.627	1280	31.4
0.000	0.026	0.265	0.709	1022	27.3
0.000	0.028	0.213	0.760	977	27.1
0.000	0.032	0.197	0.776	869	24.3
0.000	0.039	0.187	0.785	691	20.1
0.000	0.041	0.161	0.811	677	19.8
0.000	0.041	0.154	0.817	659	19.9
0.000	0.043	0.121	0.849	630	19.7
0.000	0.053	0.075	0.889	441	16.8
0.000	0.055	0.061	0.901	407	16.4
0.000	0.125	0.012	0.919	93	7.4
0.000	0.152	0.000	0.899	50	5.9
[COC ₂ mPIP][FAP]					
0.000	0.000	0.931	0.000	–	–
0.000	0.005	0.821	0.110	317	22.0
0.000	0.007	0.695	0.241	534	34.4
0.000	0.009	0.624	0.318	604	35.3
0.000	0.009	0.598	0.345	666	38.3
0.000	0.013	0.478	0.470	686	36.2
0.000	0.016	0.427	0.524	658	32.8
0.000	0.018	0.414	0.538	611	29.9
0.000	0.020	0.360	0.593	618	29.7
0.000	0.027	0.251	0.705	577	26.1
0.000	0.030	0.217	0.739	543	24.6
0.000	0.032	0.188	0.768	528	24.0
0.000	0.035	0.165	0.791	496	22.6
0.000	0.042	0.156	0.801	425	19.1
0.000	0.045	0.135	0.822	406	18.3
0.000	0.070	0.071	0.885	267	12.6
0.000	0.100	0.036	0.912	158	9.1
0.000	0.152	0.000	0.899	50	5.9
[COC ₂ mPYR][FAP]					
0.000	0.000	0.946	0.000	–	–
0.000	0.004	0.855	0.089	396	22.3
0.000	0.007	0.708	0.239	640	34.1
0.000	0.008	0.665	0.284	691	35.5
0.000	0.008	0.662	0.287	698	35.9
0.000	0.011	0.559	0.393	736	35.7
0.000	0.011	0.535	0.418	800	38.0
0.000	0.013	0.496	0.459	774	35.3
0.000	0.014	0.483	0.472	739	33.7
0.000	0.014	0.475	0.480	751	34.3
0.000	0.014	0.478	0.477	747	34.1
0.000	0.017	0.435	0.521	685	30.6
0.000	0.021	0.388	0.570	633	27.1
0.000	0.027	0.299	0.661	596	24.5
0.000	0.032	0.281	0.680	527	21.3
0.000	0.034	0.251	0.711	532	20.9
0.000	0.036	0.246	0.716	505	19.9
0.000	0.043	0.190	0.772	452	18.0
0.000	0.047	0.134	0.827	430	17.6
0.000	0.056	0.056	0.901	353	16.1
0.000	0.105	0.028	0.918	145	8.7
0.000	0.152	0.000	0.899	50	5.9

^a Standard uncertainties u are $u(x) < 0.002$, $u(T) = 0.02$ K.

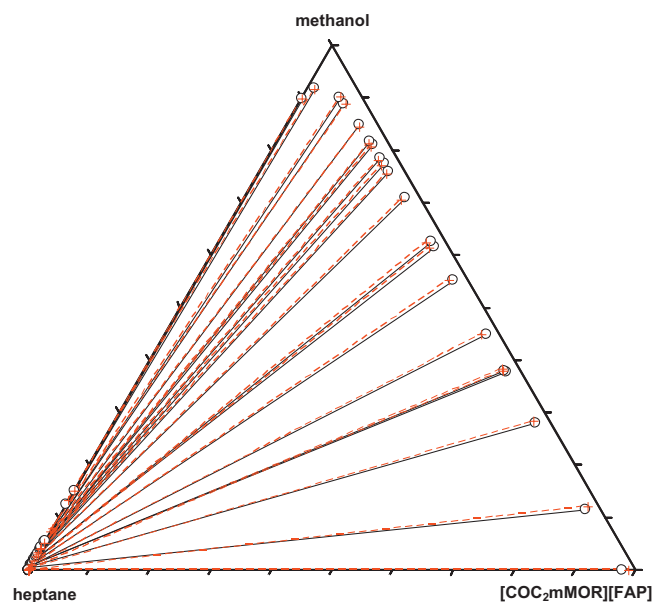


Fig. 1. Experimental versus NRTL correlations for the composition tie lines of the ternary system {[COC₂mMOR][FAP] (1) + methanol (2) + heptane (3)} at 298.15 K; (O–O) experimental data, (+–+) NRTL correlation.

easy to see that results calculated from NRTL correlation are very good fitted to experimental ones. Figs. 1S–3S presents comparison of obtained data with previously studied ILs based on [NTf₂][–] anion. In each case the miscibility gap is lower for [FAP][–] based ILs. Selectivity and solute distribution ratio for all six ILs as function of methanol mole fraction in the hydrocarbon-rich phase can be seen in Figs. 4 and 5, respectively.

In each case S and β increase at low concentration of solute and gain the maximum value in the range of x_2^{HC} from 0.009 to 0.013 and next decrease with increase of methanol concentrations.

The selectivity is the highest for [COC₂mMOR][FAP] from [FAP][–] based ILs and it is three times higher than for [COC₂mPIP][FAP] and [COC₂mPYR][FAP] in low concentration of solute. For this

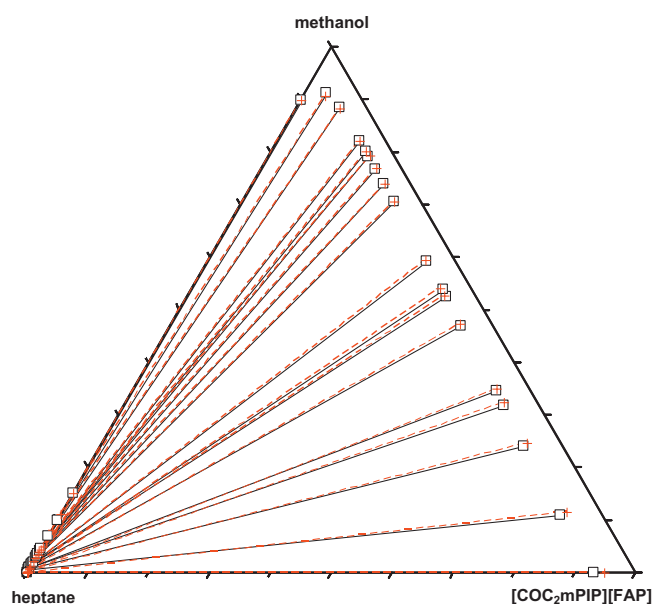


Fig. 2. Experimental versus NRTL correlations for the composition tie lines of the ternary system {[COC₂mPIP][FAP] (1) + methanol (2) + heptane (3)} at 298.15 K; (□–□) experimental data, (+–+) NRTL correlation.

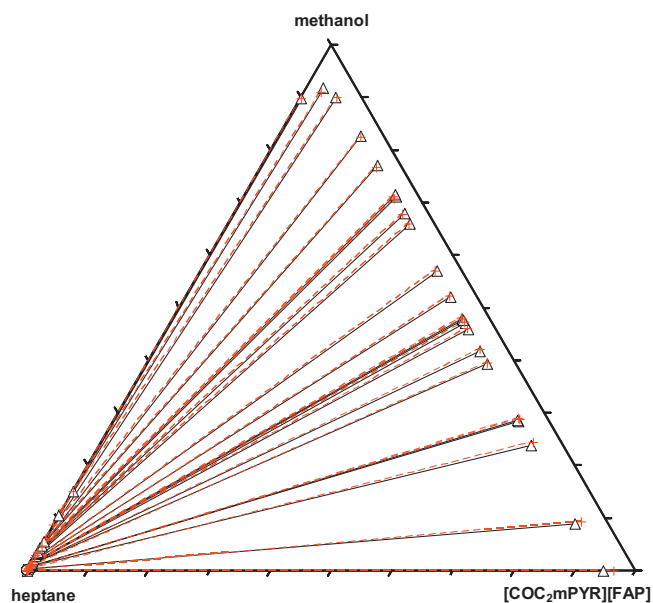


Fig. 3. Experimental versus NRTL correlations for the composition tie lines of the ternary system {[COC₂mPYR][FAP] (1)+methanol (2)+heptane (3)} at 298.15 K; (Δ — Δ) experimental data, (+—) NRTL correlation.

two ILs values of S are comparable due to similar cation structures. The same effect was observed previously for ILs based on [NTf₂][−] anion. ILs based on [FAP][−] anion indicate higher selectivities in relation to respective [FAP][−] ILs with the same cations. However [COC₂mMOR][FAP] has better S than [COC₂mPIP][NTf₂] and [COC₂mPYR][NTf₂]. In higher concentration of methanol the selectivities reach the similar values for all ILs.

The solute distribution ratio is also comparable for [COC₂mPIP][FAP] and [COC₂mPYR][FAP] and the highest one is for [COC₂mMOR][FAP]. The differences in respect to the behavior of selectivity are similar values of β for ILs based on [FAP][−] and [NTf₂][−] anions.

High values of selectivities and solute distribution ratios for investigated ILs indicate that they can be used successfully in separation of methanol/heptane azeotropic mixture, especially [COC₂mMOR][FAP]. Nevertheless studied previously ILs based on [NTf₂][−] anion reveal better properties to realize this process.

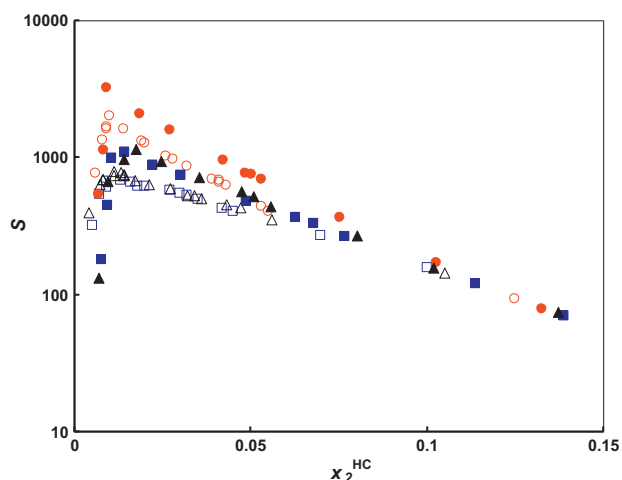


Fig. 4. Selectivity, S as a function of the mole fraction of methanol in the hydrocarbon-rich phase for the investigated systems; ($\circ$) [COC₂mMOR][FAP], ($\square$) [COC₂mPIP][FAP], (Δ) [COC₂mPYR][FAP], ($\bullet$) [COC₂mMOR][NTf₂] [3], ($\blacksquare$) [COC₂mPIP][NTf₂] [3], ($\blacktriangle$) [COC₂mPYR][NTf₂] [3].

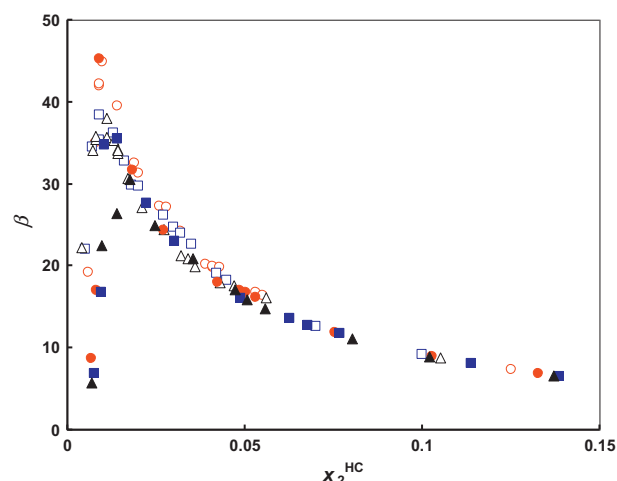


Fig. 5. Solute distribution ratio, β as a function of the mole fraction of methanol in the hydrocarbon-rich phase for the investigated systems; ($\circ$) [COC₂mMOR][FAP], ($\square$) [COC₂mPIP][FAP], (Δ) [COC₂mPYR][FAP], ($\bullet$) [COC₂mMOR][NTf₂] [3], ($\blacksquare$) [COC₂mPIP][NTf₂] [3], ($\blacktriangle$) [COC₂mPYR][NTf₂] [3].

5. Conclusions

An experimental investigation on liquid–liquid equilibria of the ternary mixtures composed of three ionic liquids based on trifluorotris(perfluoroethyl)phosphate anion + methanol + heptane has been carried out at $T = 298.15$ K. It is fact that presented ionic liquids could perform a role of solvent in liquid–liquid extraction, particularly 4-(2-methoxyethyl)-4-methylmorpholinium trifluorotris(perfluoroethyl)phosphate. The results were correlated with the NRTL equation providing a very good fit of the experimental results.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fluid.2012.11.014>.

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